

Node-RED sẽ **publish** các topics này để gửi commands đến ESP32:

1. `doorbell/cmd/snapshot`

Frontend yêu cầu chụp ảnh

Node-RED publish:

```
1
2  {
3
4    "command_id": "cmd_1735490600_snapshot",
5
6    "timestamp": 1735490600
7
8  }
9
```

2. `doorbell/cmd/speak`

Frontend gửi audio message để ESP32 phát

Node-RED publish:

```
1
2  {
3
4    "command_id": "cmd_1735490700_speak",
5
6    "audio_url": "https://xxx.supabase.co/.../message.wav",
7
8    "volume": 80,
9
10   "timestamp": 1735490700
11
12  }
13
```

3. doorbell/cmd/siren

Frontend bật/tắt còi

Node-RED publish:

```
1
2 {
3
4   "command_id": "cmd_1735490800_siren",
5
6   "action": "on",
7
8   "duration": 5,
9
10  "timestamp": 1735490800
11
12 }
13
```

4. doorbell/cmd/settings

Node-RED notify ESP32 khi settings thay đổi

Node-RED publish:

```
1
2 {
3
4   "speaker_volume": 80,
5
6   "pir_enabled": true,
7
8   "alarm_enabled": false,
9
10  "timestamp": 1735490900
11
12 }
13
```

